

Software Requirement Specification - IntelliCV

Hardware Requirements

- **Processor:** Minimum Intel i5 / AMD Ryzen 5 or equivalent — to support PDF parsing, backend processing, and AI inference requests.
 - **RAM:** At least 8 GB (16 GB recommended) for handling large PDF files, LLM prompts, and database operations efficiently.
 - **Storage:** Minimum 10–20 GB free space for uploaded documents, extracted text, logs, and database storage.
 - **Display:** Minimum 1366×768 resolution for comfortable use of development tools and dashboards.
 - **System Type:** 64-bit operating system for compatibility with Node.js, Python, and database drivers.
 - **Connectivity:** Stable internet connection required for Gemini API calls, dependency installation, and cloud services.
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Software Requirements

- **Operating System:**
 - Windows 10 / 11
 - Linux (Ubuntu 20.04+)
 - macOS (Intel / Apple Silicon)
- **Backend Runtime:**
 - Node.js 18+ — backend API, authentication, database interaction
 - Python 3.9+ — document processing and LLM integration
- **Database Systems:**
 - MySQL 8.x — structured relational data (users, documents, projects, resumes)
 - Optional: MongoDB / Vector DB (FAISS, ChromaDB) — embeddings and semantic search
- **Frameworks & Libraries:**
 - Express.js — REST API development
 - Sequelize ORM — database abstraction and query management
 - PyPDF2 — PDF text extraction
 - google-generativeai — Gemini LLM integration

- dotenv — environment variable management
 - **Editor / IDE:**
 - Visual Studio Code — development and debugging
 - **Version Control:**
 - Git / GitHub — source control, collaboration, and backups
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Functional Requirements

1) User Authentication

Users should be able to register and log in using email and password.

The system shall securely store passwords using hashing and issue JWT tokens for authenticated sessions.

2) Document Upload

Users should be able to upload PDF documents such as resumes, certificates, and project reports. The system shall associate each uploaded document with the authenticated user.

3) Text Extraction

The system shall extract readable text from uploaded PDF files automatically.

Extracted text shall be stored temporarily or persistently for downstream processing.

4) AI-Based Content Analysis

The system shall analyze extracted document content using a Large Language Model (Gemini).

The analysis shall identify relevant information such as projects, skills, certifications, and experience.

5) Structured Query Generation

The system shall generate Sequelize-compatible INSERT queries based on analyzed content.

Each generated query shall be categorized by type (Document, Project, Certificate, etc.).

6) Data Storage

The system shall store structured information in the MySQL database under appropriate entities.

The system shall optionally store embeddings for semantic retrieval and resume matching.

7) Resume Generation

Users should be able to upload a job description.

The system shall generate a tailored resume using AI based on user data and job requirements.